QuickPhone — Physical ERD (complete): 16 tables with all keys and attributes

One diagram for the whole data model. Each line is named both ways: top or left from the "one" side, bottom or right back from the "many" side. PK primary key · FK foreign key (→ table) · UK unique · CK checked value list · NN not null · derived = denormalized

store — D7 Staff & Stores

PK	**store_id**	INTEGER	NN
UK	store_name	VARCHAR(60)	NN
	address	VARCHAR(200)	NN
	phone	VARCHAR(20)	NN
	email_address	VARCHAR(120)	NN
	business_hours	VARCHAR(120)	

appointment — D2 Appointments

PK	**appointment_id**	INTEGER	NN
FK	customer_id	INTEGER	→ *customer*
FK	store_id	INTEGER	→ *store*
FK	technician_id	INTEGER	→ *employee*
	scheduled_at	TIMESTAMP	NN
CK	appt_status	VARCHAR(12)	NN

repair_job — D3 Repair Jobs

PK	**repair_job_id**	INTEGER	NN
UK	ticket_number	VARCHAR(12)	NN
FK	device_id	INTEGER	→ *device*
FK	store_id	INTEGER	→ *store*
FK,UK	appointment_id	INTEGER	→ *appointment*
FK	assigned_tech_id	INTEGER	→ *employee*
CK	status	VARCHAR(20)	derived · NN
	reported_issue	VARCHAR(500)	NN
	intake_condition	VARCHAR(500)	
	diagnosis_notes	VARCHAR(2000)	
	repair_notes	VARCHAR(2000)	
	check_in_at	TIMESTAMP	NN
	completed_at	TIMESTAMP	
	final_cost	DECIMAL(10,2)	
	email_sent_at	TIMESTAMP	
FK	email_sent_by_id	INTEGER	→ *employee*
CK	email_status	VARCHAR(10)	
	picked_up_at	TIMESTAMP	
	id_verified_flag	BOOLEAN	NN
	signature_file	VARCHAR(255)	

payment — D3 Repair Jobs

PK	**payment_id**	INTEGER	NN
FK	repair_job_id	INTEGER	→ *repair_job*
	amount	DECIMAL(10,2)	NN
CK	method	VARCHAR(12)	NN
CK	payment_status	VARCHAR(10)	NN
	paid_at	TIMESTAMP	NN

employee — D7 Staff & Stores

PK	**employee_id**	INTEGER	NN
FK	store_id	INTEGER	→ *store*
	first_name	VARCHAR(40)	NN
	last_name	VARCHAR(40)	NN
UK	email	VARCHAR(120)	NN
CK	role	VARCHAR(12)	NN
	password_hash	VARCHAR(255)	NN
	active_flag	BOOLEAN	NN

customer — D1 Customers & Devices

PK	**customer_id**	INTEGER	NN
	first_name	VARCHAR(40)	NN
	last_name	VARCHAR(40)	NN
IX	phone	VARCHAR(20)	NN
	email	VARCHAR(120)	
	created_at	TIMESTAMP	NN

job_part — D3 Repair Jobs

PK	**job_part_id**	INTEGER	NN
FK	repair_job_id	INTEGER	→ *repair_job*
FK	part_id	INTEGER	→ *part*
FK	purchase_order_id	INTEGER	→ *purchase_order*
	quantity	INTEGER	NN
CK	part_status	VARCHAR(8)	NN
	recorded_at	TIMESTAMP	NN

device — D1 Customers & Devices

PK	**device_id**	INTEGER	NN
FK	customer_id	INTEGER	→ *customer*
	device_type	VARCHAR(20)	NN
	brand	VARCHAR(40)	NN
	model	VARCHAR(60)	NN
IX	serial_or_imei	VARCHAR(40)	

job_status_history — D3 Repair Jobs

PK	**history_id**	BIGINT	NN
FK	repair_job_id	INTEGER	→ *repair_job*
CK	status	VARCHAR(20)	NN
FK	changed_by_id	INTEGER	→ *employee*
	changed_at	TIMESTAMP	NN
	note	VARCHAR(300)	

part — D4 Parts Inventory

PK	**part_id**	INTEGER	NN
FK	default_supplier_id	INTEGER	→ *supplier*
	part_name	VARCHAR(100)	NN
	device_model	VARCHAR(60)	
	grade	VARCHAR(20)	
	standard_cost	DECIMAL(10,2)	NN
	active_flag	BOOLEAN	NN

supplier — D6 Suppliers

PK	**supplier_id**	INTEGER	NN
UK	supplier_name	VARCHAR(100)	NN
	contact_name	VARCHAR(80)	
	phone	VARCHAR(20)	
	email	VARCHAR(120)	
	active_flag	BOOLEAN	NN

store_part_stock — D4 Parts Inventory

PK,FK	**store_id**	INTEGER	→ *store*
PK,FK	**part_id**	INTEGER	→ *part*
CK	quantity_on_hand	INTEGER	derived · NN
	reorder_point	INTEGER	NN

quote — D3 Repair Jobs

PK	**quote_id**	INTEGER	NN
FK	repair_job_id	INTEGER	→ *repair_job*
	labor_amount	DECIMAL(10,2)	NN
	parts_amount	DECIMAL(10,2)	NN
	total_amount	DECIMAL(10,2)	derived · NN
CK	decision	VARCHAR(10)	NN
	decision_at	TIMESTAMP	
FK	decision_by_id	INTEGER	→ *employee*

inventory_txn — D4 Parts Inventory

PK	**inventory_txn_id**	BIGINT	NN
FK	store_id	INTEGER	→ *store_part_stock*
FK	part_id	INTEGER	→ *store_part_stock*
FK	recorded_by_id	INTEGER	→ *employee*
CK	txn_type	VARCHAR(12)	NN
	quantity_change	INTEGER	NN
	reason	VARCHAR(200)	
	txn_at	TIMESTAMP	NN

purchase_order — D5 Purchase Orders

PK	**purchase_order_id**	INTEGER	NN
FK	store_id	INTEGER	→ *store*
FK	supplier_id	INTEGER	→ *supplier*
FK	created_by_id	INTEGER	→ *employee*
FK	approved_by_id	INTEGER	→ *employee*
CK	po_status	VARCHAR(20)	NN
	created_at	TIMESTAMP	NN
	approved_at	TIMESTAMP	
	placed_at	TIMESTAMP	
	rejection_reason	VARCHAR(300)	

po_line — D5 Purchase Orders

PK	**po_line_id**	INTEGER	NN
FK	purchase_order_id	INTEGER	→ *purchase_order*
FK	part_id	INTEGER	→ *part*
	quantity_ordered	INTEGER	NN
	quantity_received	INTEGER	NN
	unit_cost	DECIMAL(10,2)	NN

schedules / is scheduled at — store ↔ appointment

employs / is employed by — store ↔ employee

handles / is handled by — employee ↔ appointment

books / is booked by — customer ↔ appointment

leads to / results from — appointment ↔ repair_job

works on / is assigned to — employee ↔ repair_job

sends notification for / has notification sent by — employee ↔ repair_job

receives / is received for — repair_job ↔ payment

uses / is used on — repair_job ↔ job_part

is included in / includes — job_part ↔ part

owns / is owned by — customer ↔ device

undergoes / is performed on — device ↔ repair_job

tracks / belongs to — repair_job ↔ job_status_history

generates / is generated for — employee ↔ job_status_history

appears in / is stock of — part ↔ store_part_stock

logs / is logged for — store_part_stock ↔ inventory_txn

creates / is created by — employee ↔ purchase_order

fulfills / is fulfilled by — supplier ↔ purchase_order

approves / is approved by — employee ↔ purchase_order

provides / is provided by — supplier ↔ part

changes / is made by — employee ↔ job_status_history

decides / is decided by — employee ↔ quote

contains / is contained in — purchase_order ↔ po_line

records / is recorded by — employee ↔ inventory_txn

appears on / lists — part ↔ po_line

supplies / is supplied by — supplier ↔ po_line

Every foreign key is drawn as a relationship line. EMPLOYEE plays several roles: the technician assigned a job, the sender of the pickup notification, and the person who changes a status, decides a quote, records a stock change, or creates or approves a purchase order.